

Mahmoud Alyosify

Specialised AI Agent Systems Engineer | Multi-Agent Orchestration, Tool Use & Workflow Automation | MSc Artificial Intelligence, Queen's University (Canada)

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PROFESSIONAL SUMMARY

AI Agent Systems Engineer building **autonomous, multi-step systems that do real work** — not chat wrappers. Architected **HORUS Sentinel**, a swarm of specialised agents that continuously collect open-source intelligence and correlate every finding into a **living knowledge graph**, over which a self-hosted fine-tuned LLM produces **evidence-backed reports**, compressing hours of analyst work into minutes. Fluent in **LLM API integration** (prompt design, **structured outputs**, **tool / function calling**), **RAG**, and end-to-end agentic automation pipelines. Also studies agents empirically — a GitHub-mining project quantifying **review effort and failure modes of agent-generated pull requests**. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, researching token efficiency — the cost dimension that decides whether agent loops are economically viable.

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Nov 2026
Expected Nov 2026
Queen's University, Kingston, Ontario, Canada

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines.**

B.Sc. Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
GPA: 3.53 / 4.00
Assiut University, Egypt

- Graduation Project: **Vitalism Solution** — **Grade A+; Semi-Finalist, Microsoft Imagine Cup 2023.**

TECHNICAL SKILLS

Agentic Systems	Multi-Agent Architectures, Agent Orchestration, Think–Plan–Act Loops, Tool / Function Calling, Structured Outputs, Autonomous Task Decomposition, Workflow Automation, LangChain, Knowledge-Graph Grounding
LLM Integration	LLM APIs (Anthropic, OpenAI), Prompt Engineering, RAG, Vector Search (FAISS, ChromaDB), Fine-Tuning (LoRA, QLoRA, PEFT), Self-Hosted Inference, Hugging Face Transformers
Agentic Coding Tools	Claude Code, Cursor, GitHub Copilot
Engineering	Python, TypeScript, FastAPI, Docker, AWS, ONNX, CI/CD, MongoDB, MySQL, Git/GitHub, Linux
Evaluation	LLM-as-Judge, BERTScore, ROUGE, Token-Cost & Latency Profiling, Statistical Testing, Reproducible Experiment Design
Programming	Python, TypeScript, JavaScript, SQL, C++, C#, C, Go, Bash

SELECTED AGENTIC AI PROJECTS

HORUS Sentinel — Autonomous Multi-Agent OSINT & Threat-Intelligence Platform 2026
Python · Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · Self-Hosted Inference

- Architected an autonomous intelligence analyst: a **swarm of specialised passive agents**, each owning one collection domain, runs continuously and independently rather than under a single monolithic prompt.
- Designed a **living Intelligence Knowledge Graph** as the shared memory layer, so every agent's finding is correlated with prior evidence instead of being lost between runs.
- A **self-hosted, fine-tuned language model** reasons over that graph to deliver **prioritised, evidence-backed reports** — every conclusion traceable to its source — compressing hours of manual analyst work into minutes.

Automated Multimodal Agent — PDF to Narrated Presentation 2026
Agentic Orchestration · Multimodal Generation · Python

- Built an autonomous **think–plan–act** pipeline that converts a source PDF into a fully narrated PowerPoint deck, coordinating document understanding, structured slide generation and audio narration as distinct tool-using stages.
- Removed an entirely manual workflow end to end — the agent decides structure and content, not just formatting.

Analyzing Review Effort: Human vs. Agentic Pull Requests 2026
Empirical Study · GitHub Mining · Agent Evaluation

- Built a reproducible GitHub-mining pipeline quantifying **review effort and recurring failure modes** of AI-agent-generated pull requests versus human-authored ones across large open-source repositories — evidence about where coding agents actually break down.

Verbosity-Aware Decoding — Token Economics for Agent Loops 2026 – Present
MSc Research Project (CISC 898), Queen's University · PyTorch · AWS GPU

- Designed a **training-free LogitsProcessor** that cuts generated tokens at decoding time under an **entropy-driven gate** — directly addressing the cost multiplier that makes long multi-agent loops expensive — validated across three model families with latency and GPU-energy measurement.

Additional Projects

- **Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** GTD/GDELT records and deployed it on **AWS** as an interactive assistant.
- **SimCLR-Vision-SSL** — contrastive representation learning with ONNX/FAISS retrieval — the vector-search layer behind RAG-style agent memory.
- **Also:** SCRAP · AudioShield · RL-Cloud-Autoscaler (PPO/DQN) · Vitalism Solution (rPPG).

PROFESSIONAL EXPERIENCE

Machine Learning Instructor <i>National Telecommunication Institute (NTI)</i>	<i>Jul 2025 – Sep 2025</i> Cairo, Egypt
• Delivered a 90-hour hands-on ML program covering supervised learning, PCA and neural networks with practical labs.	
AI in Cybersecurity Intern <i>e& Egypt — via NTI (On-the-Job Training)</i>	<i>Jun 2025 – Jul 2025</i> Cairo, Egypt
• Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.	
Systems Engineer (Military Service) <i>National Company for Roads Building & Development</i>	<i>Feb 2024 – Mar 2025</i> Cairo, Egypt
• Supported backend systems for national toll infrastructure serving 50%+ of Egyptian vehicles ; co-developed an internal data-management application in C#, .NET, Entity Framework, SQL Server .	
Online Instructor — AI, Algorithms & Data Science <i>Udemy & YouTube</i>	<i>2020 – Present</i> Remote
• Built Arabic-language technical curricula reaching 10,000+ learners and 700K+ YouTube views .	

CERTIFICATIONS & AWARDS

Generative AI & Prompt Engineering — IBM / Coursera	AWS Academy — ML Foundations, ML for NLP, Data Engineering
Machine Learning Specialization — DeepLearning.AI	HCIP – AI (Huawei)
Digilians Presidential Scholarship — Queen’s University	1st Place — Smart Cities Hackathon 2022
Semi-Finalist — Microsoft Imagine Cup 2023	Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.